

Selected Use Cases — Part B

System Analysis & Design — Group 17

Group No.	Organization	Submission Date	Student IDs
17	TLV Transfers	27.05.2026	_____

UC17: Send Trip Offer

The dispatcher assigns open trips to a driver region. The system ranks eligible drivers by proximity, workload, and availability, then sends a WhatsApp message with full trip details and a unique confirmation link to the top-ranked driver.

Primary Actor

Dispatcher

Preconditions

1. At least one trip exists with status "Open."
2. Dispatcher is logged in to the dispatch dashboard.
3. At least one active driver is registered for the relevant region.

Postconditions

1. A WhatsApp offer has been sent to the highest-ranked eligible driver.
2. Trip status changed to "Pending Approval."
3. System is actively monitoring for driver response.

Main Success Scenario (MSS)

1. Dispatcher selects one or more open trips from the dashboard.
2. Dispatcher assigns the trips to a driver region or group.
3. System loads all active drivers registered for that region.
4. System filters drivers by vehicle type and passenger capacity.
5. System ranks the filtered drivers by proximity, current workload, and availability.
6. System packages the selected trips for the top-ranked driver, verifying there are no time conflicts between trips.
7. System sends a WhatsApp message to the top-ranked driver containing full trip details and a unique confirmation link.
8. Trip status is updated to "Pending Approval."
9. System begins monitoring for driver response.

Alternative Flows / Extensions

- 4a. No eligible drivers found → System displays "No available driver" alert; Dispatcher may reassign to a supplier instead.

- 7a. WhatsApp delivery fails → System retries; if retry fails, Dispatcher receives an alert to handle manually.
- 9a. Driver does not respond within the allowed time → System automatically forwards the offer to the next driver in the ranked list.

Notes

- Dispatcher may cancel the assignment at any point before the driver responds; trip returns to "Open."
- If all ranked drivers are exhausted without acceptance, the trip is flagged "Unassigned" and Dispatcher is notified.

Assign Trip to Region

The dispatcher assigns an open trip to a geographic region, making it visible in the dispatch queue for that region and enabling driver matching to begin.

Primary Actor

Dispatcher

Preconditions

1. A trip exists in the system with status "Open."
2. At least one active Region is configured in the system.
3. Dispatcher is logged in to the dispatch dashboard.

Postconditions

1. Trip is associated with the selected region.
2. Trip is visible in the region's dispatch queue.
3. Trip remains with status "Open," ready for driver assignment.

Main Success Scenario (MSS)

1. Dispatcher views the open trips panel in the dashboard.
2. Dispatcher drags a trip onto a region or selects a region from the dropdown.
3. System validates that the selected region exists and is active.
4. System updates the Trip's region field.
5. System displays the trip under the selected region in the dispatch queue.
6. Use case ends.

Alternative Flows / Extensions

- 3a. Region is inactive → System shows an error message; trip remains unassigned.
- 2a. Dispatcher selects a supplier instead of a region → System routes to the supplier dispatch path.

UC07: Respond to Trip Offer

The driver receives a WhatsApp message with a unique confirmation link, opens the mobile approval page, and accepts or rejects each offered trip individually. Accepted trips are confirmed and Ride Control is updated; rejected trips are forwarded to the next driver.

Primary Actor

External Driver - Supplier

Preconditions

1. Driver received a WhatsApp message containing a confirmation link.
2. The confirmation link is still valid.
3. The offered trip(s) are still in "Pending Approval" status.

Postconditions

1. Accepted trips are updated to status "Assigned."
2. Ride Control System is updated for each accepted trip.
3. Driver received a WhatsApp confirmation for accepted trips.
4. Rejected trips are forwarded to the next available driver in the ranked list.

Main Success Scenario (MSS)

1. Driver receives a WhatsApp message with trip details and a confirmation link.
2. Driver opens the confirmation link.
3. System validates the link.
4. System verifies the offered trips are still pending and have not been assigned to another driver.
5. Confirmation page loads, displaying each offered trip with: time, pickup address, destination, and number of passengers.
6. Driver reviews each trip individually and selects "Accept" or "Cannot" for each one.
7. System saves the driver's selections.
8. For each accepted trip: status is updated to "Assigned" and Ride Control is updated (triggers Update Ride Control).
9. Driver receives a WhatsApp confirmation: "Confirmed! Thank you."
10. For each rejected trip: offer is automatically forwarded to the next driver in the ranked list.

Alternative Flows / Extensions

- 3a. Link is invalid → Page displays "Invalid link" message; session ends.
- 3b. Link has expired → Page displays "This link is no longer active" message; session ends.
- 4a. Trip already assigned to another driver → Page displays "This offer is no longer available."

- 6a. Driver rejects all trips → All trips forwarded to next driver; Dispatcher notified if no drivers remain. «Extend» Forward Offer to Next Driver.

Notes

- Driver may accept some trips and reject others in the same session; each decision is handled independently.

Forward Offer to Next Driver

When a driver rejects an offer or an offer times out, the system automatically re-queues the trip and sends a new offer to the next eligible driver in the ranked list. If no drivers remain, the trip is escalated to the manager.

Primary Actor

External Driver - Supplier

Preconditions

1. A driver has rejected a trip offer, or an offer has timed out.
2. The trip still has status "Open" or "Pending Approval."
3. The system has a ranked list of drivers for the trip's region.

Postconditions

1. Trip is re-offered to the next eligible driver, OR
2. Trip is escalated to the manager if no eligible drivers remain.

Main Success Scenario (MSS)

1. Driver rejects the trip offer (via WhatsApp reply or Decline button on confirmation page).
2. System sets the current Offer status to "Rejected" and Trip status back to "Open."
3. System queries remaining ranked drivers in the region, excluding drivers who already received an offer for this trip.
4. System selects the next highest-ranked eligible driver.
5. System creates a new Offer and generates a unique confirmation URL for driver approval.
6. System sends a new WhatsApp offer message to the next driver.
7. Use case ends.

Alternative Flows / Extensions

- 3a. No eligible drivers remain → System sets Trip to "Unassigned," sends escalation WhatsApp to manager, use case ends.

UC08: Update Ride Control

Following a driver's acceptance of a trip, the system automatically synchronizes the assignment with the external Ride Control system, updating it with the driver's identity and vehicle details, ensuring both systems reflect the confirmed assignment.

Primary Actor

Ride Control «System»

Preconditions

1. A driver has accepted one or more trips.
2. Trip status in the dispatch system is "Assigned."
3. Ride Control System is available and reachable.

Postconditions

1. Ride Control System reflects the assigned driver for each trip.
2. Trip records in Ride Control are updated with driver name, phone number, and vehicle details.
3. Both systems are fully synchronized on the trip assignment.

Main Success Scenario (MSS)

1. Driver confirms acceptance of a trip (via Respond to Trip Offer).
2. System automatically initiates an update request to Ride Control — no manual action required.
3. System transmits driver assignment details to Ride Control System: driver name, phone number, and vehicle information.
4. Ride Control System processes the update and confirms receipt.
5. Trip record in both systems reflects status "Assigned" with full driver details.
6. System logs the successful synchronization with a timestamp.

Alternative Flows / Extensions

- 3a. Ride Control System is temporarily unavailable → System retries; if retries exhausted, Dispatcher receives an alert to verify manually.
- 4a. Ride Control returns an error response → System logs the error and alerts Dispatcher to manually confirm inside Ride Control.
- 4b. Duplicate assignment detected in Ride Control → System blocks the overwrite and alerts Dispatcher to resolve the conflict.

Notes

- This UC is triggered automatically following every successful driver acceptance and requires no input from the Dispatcher.

- If synchronization fails, the trip remains "Assigned" in the dispatch system but must be manually verified in Ride Control.